

RNA-seqの解析パイプライン: 基礎

RNA-seq Analysis Pipeline: Basics

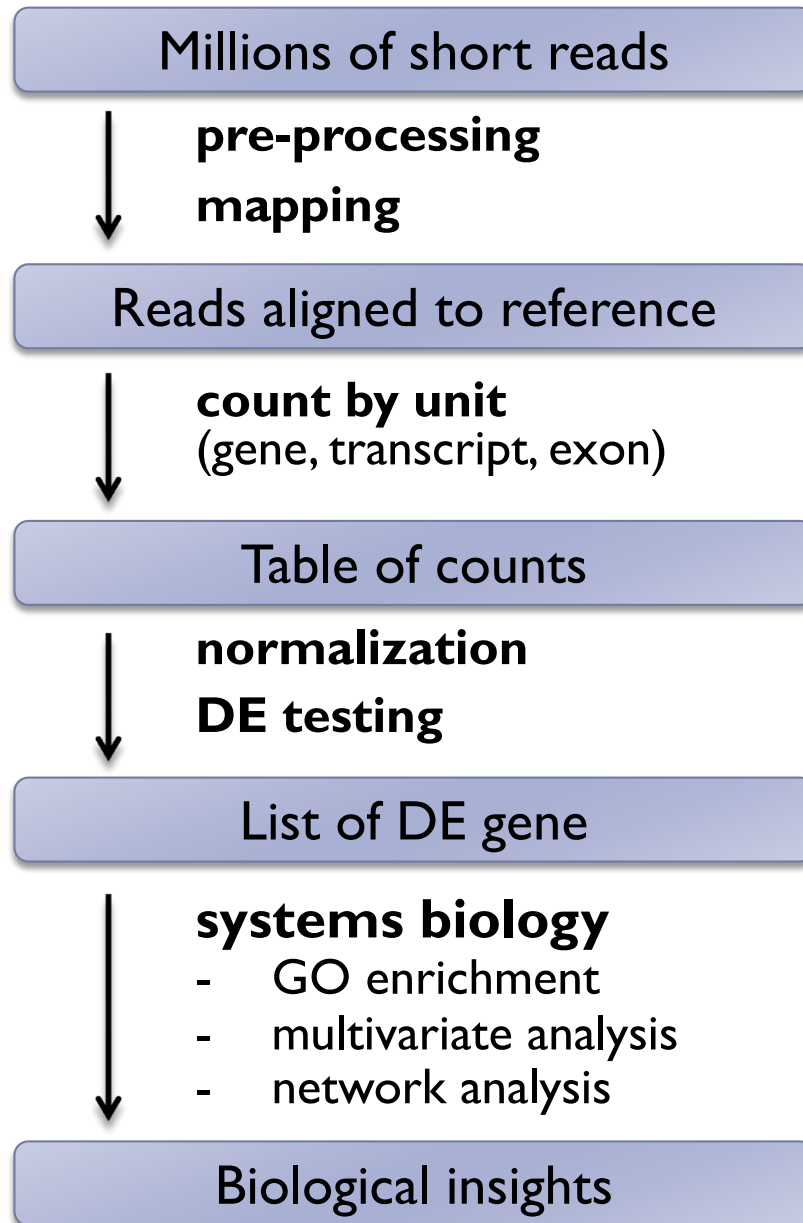
Shuji Shigenobu

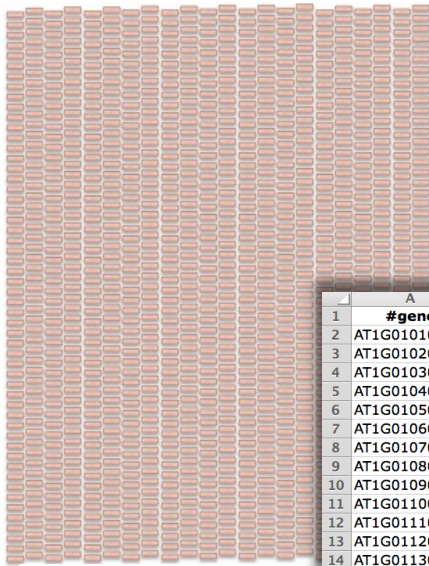
NIBB, Japan

<shige@nibb.ac.jp>

RNA-seq analysis pipeline for DE

Differential Expression analysis





NGS reads

	A	B	C	D	E	F	G
1	#gene	m1	m2	m3	h1	h2	h3
2	AT1G01010	35	77	40	46	64	60
3	AT1G01020	43	45	32	43	39	49
4	AT1G01030	16	24	26	27	35	20
5	AT1G01040	72	43	64	66	25	90
6	AT1G01050	49	78	90	67	45	60
7	AT1G01060	0	15	2	0	21	8
8	AT1G01070	16	34	6	9	20	1
9	AT1G01080	170	191	382	127	98	184
10	AT1G01090	291	346	563	171	116	453
11	AT1G01100	113	125	246	78	27	361
12	AT1G01110	0	1	1	0	0	0
13	AT1G01120	228	189	270	147	83	174
14	AT1G01130	9	11	1	0	2	2
15	AT1G01140	181	120	142	161	73	2
16	AT1G01150	0	2	0	0	0	3
17	AT1G01160	117	125	215	86	46	4
18	AT1G01170	74	57	82	36	22	5
19	AT1G01180	46	7	26	24	18	6
20	AT1G01190	0	3	2	1	2	8
21	AT1G01200	5	0	2	0	0	9
22	AT1G01210	178	203	98	205	83	10
23	AT1G01220	26	49	40	21	15	11
24	AT1G01225	4	10	6	6	0	12
25	AT1G01230	72	51	58	70	18	13
26	AT1G01240	81	89	45	62	24	14
27	AT1G01250	1	1	5	1	2	15
28	AT1G01260	15	52	37	33	27	16
29	AT1G01290	7	16	23	30	5	17
30	AT1G01300	75	115	232	89	109	18

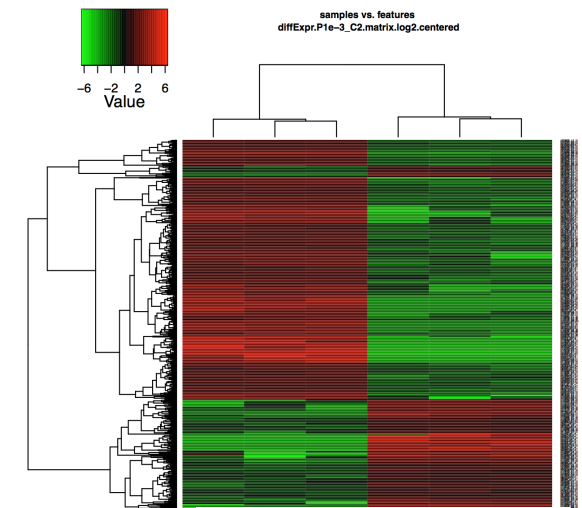
Table of counts

	A	B	C	D	E
1	gene	signal (A)	logFC (M)	p-value	FDR
2	AT5G48430	-9.17	6.30	3.7E-27	9.8E-23
3	AT2G08986	-9.15	5.04	2.8E-21	3.6E-17
4	AT5G31702	-9.74	5.68	5.0E-21	4.3E-17
5	AT1G51850	-10.07	5.32	2.2E-19	1.4E-15
6	AT2G39380	-9.82	4.97	2.9E-19	1.5E-15
7	AT2G44370	-10.23	5.48	1.1E-18	4.7E-15
8	AT3G46280	-8.19	4.93	1.3E-18	5.0E-15
9	AT3G55150	-10.44	5.86	1.8E-18	5.8E-15
10	AT2G19190	-8.70	4.55	8.5E-18	2.5E-14
11	AT1G51820	-9.40	4.40	1.3E-17	3.3E-14
12	AT2G39530	-9.16	4.36	1.7E-17	4.1E-14
13	AT1G40104	-9.58	4.61	2.2E-17	4.9E-14
14	AT2G30750	-9.97	4.64	1.5E-16	3.0E-13
15	AT2G17740	-9.12	4.18	1.8E-16	3.3E-13
16	AT2G07981	-10.07	4.75	1.0E-15	1.8E-12
17	AT1G30700	-9.21	3.91	3.5E-15	5.8E-12
18	AT5G31770	-9.12	3.76	1.9E-14	2.7E-11
19	AT1G51800	-8.47	3.99	1.9E-14	2.7E-11
20	AT1G30720	-9.06	3.75	4.7E-14	6.5E-11
21	AT4G12500	-6.58	4.39	1.5E-13	1.9E-10
22	AT1G07160	-10.76	5.02	2.0E-13	2.5E-10
23	AT2G39200	-7.57	3.98	6.0E-13	7.2E-10
24	AT2G10600	-9.30	3.50	8.8E-13	1.0E-09
25	AT1G65500	-9.60	3.50	9.6E-13	1.0E-09
26	AT5G39580	-9.24	3.45	1.3E-12	1.3E-09
27	AT5G44380	-9.51	3.45	1.7E-12	1.7E-09
28	AT5G13220	-7.20	3.88	2.8E-12	2.7E-09
29	AT2G45220	-8.45	3.54	3.9E-12	3.6E-09
30	AT1G30730	-9.80	3.45	6.0E-12	5.4E-09
31	AT1G64170	-9.91	3.39	1.2E-11	1.1E-08
32	AT4G20860	-8.10	3.49	1.5E-11	1.2E-08
33	AT4G23550	-10.08	3.49	1.5E-11	1.2E-08
34	AT4G12490	-6.75	3.88	1.6E-11	1.2E-08
35	AT5G64120	-7.09	3.76	2.1E-11	1.6E-08

List of DE genes
Table of significance score

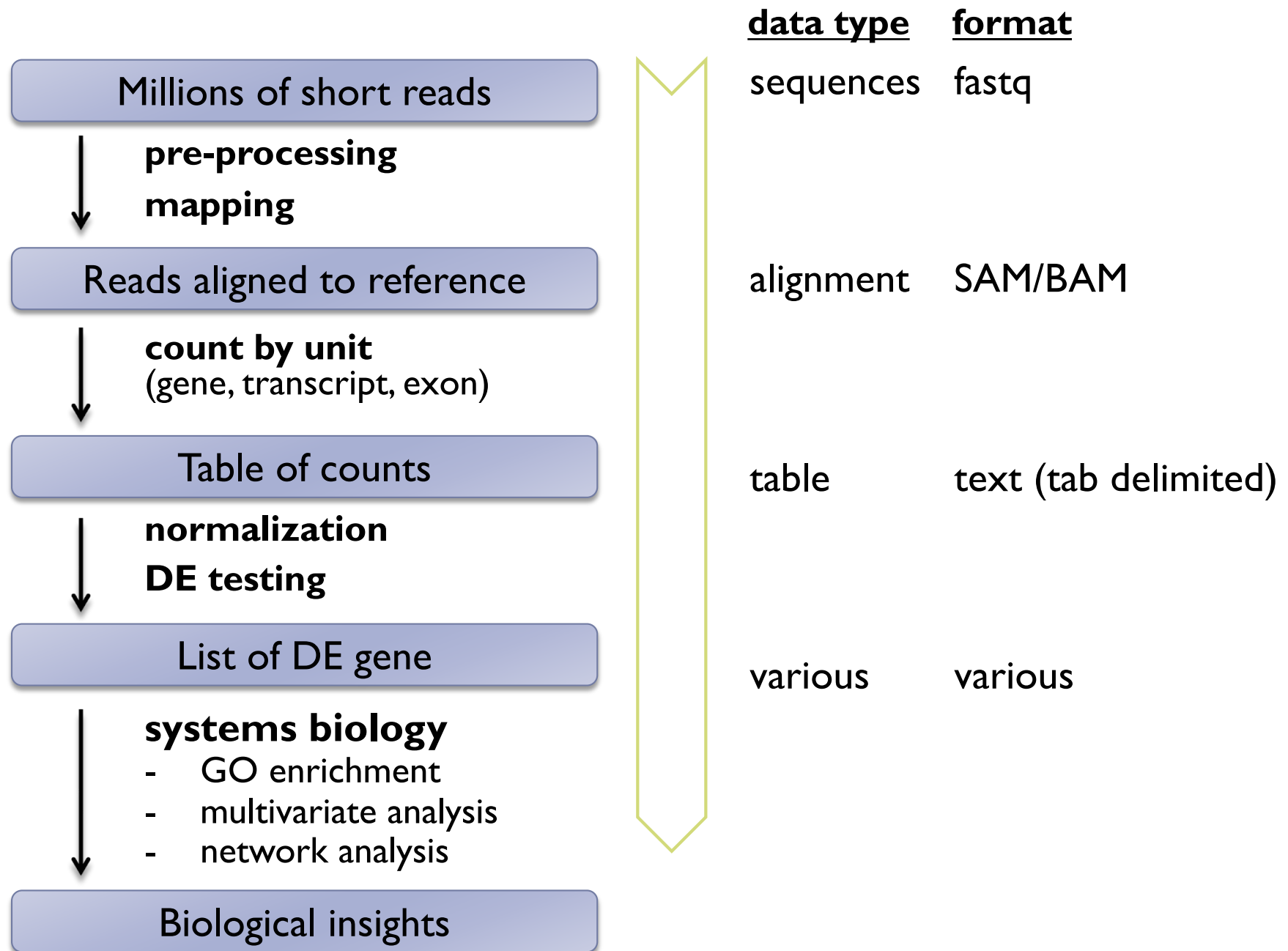


Biological insights



Systems biology:
clustering and
network analysis

RNA-seq analysis pipeline for DE



Two Basic Pipelines

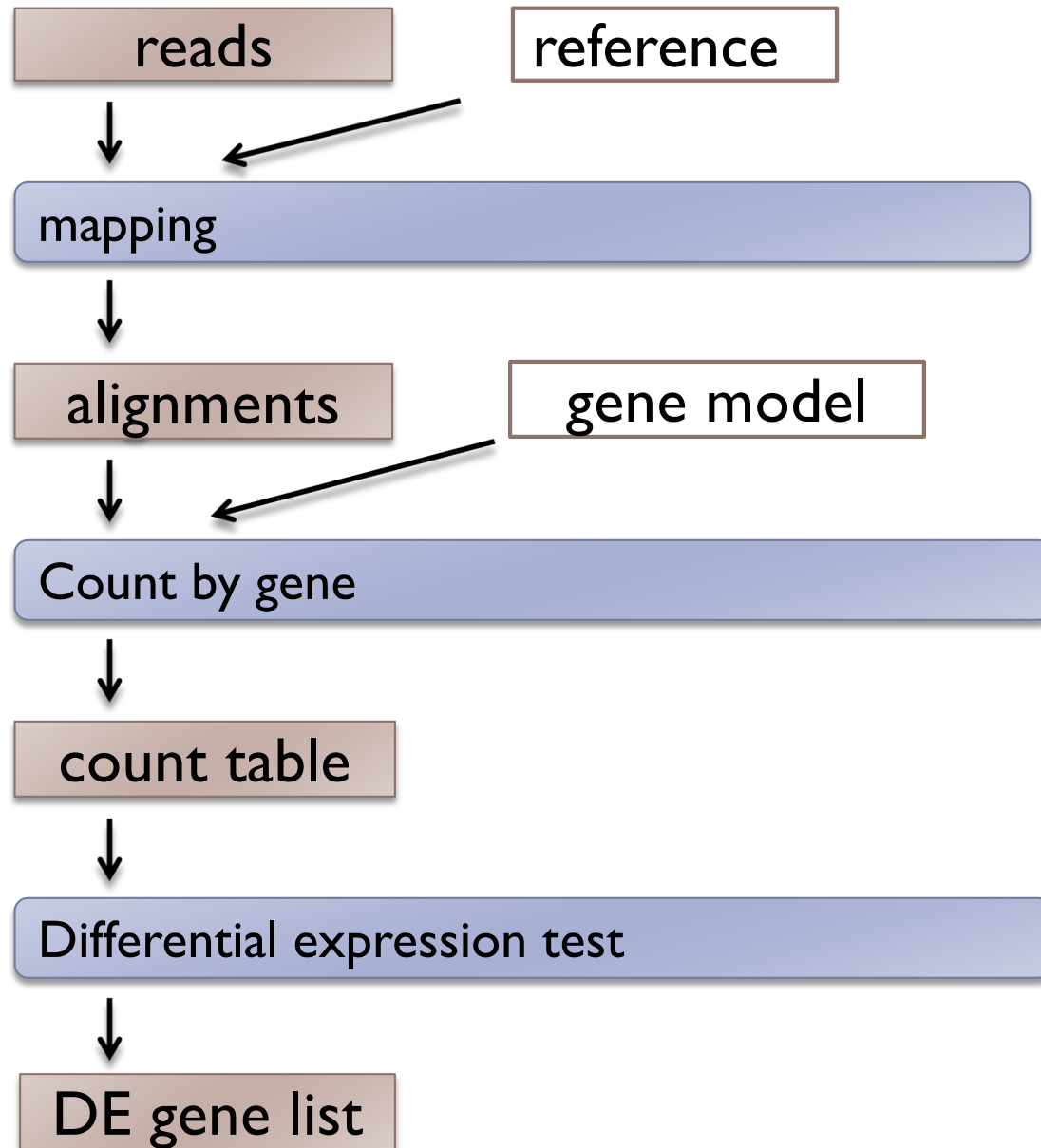
▶ Choice of reference

ゲノム参照型

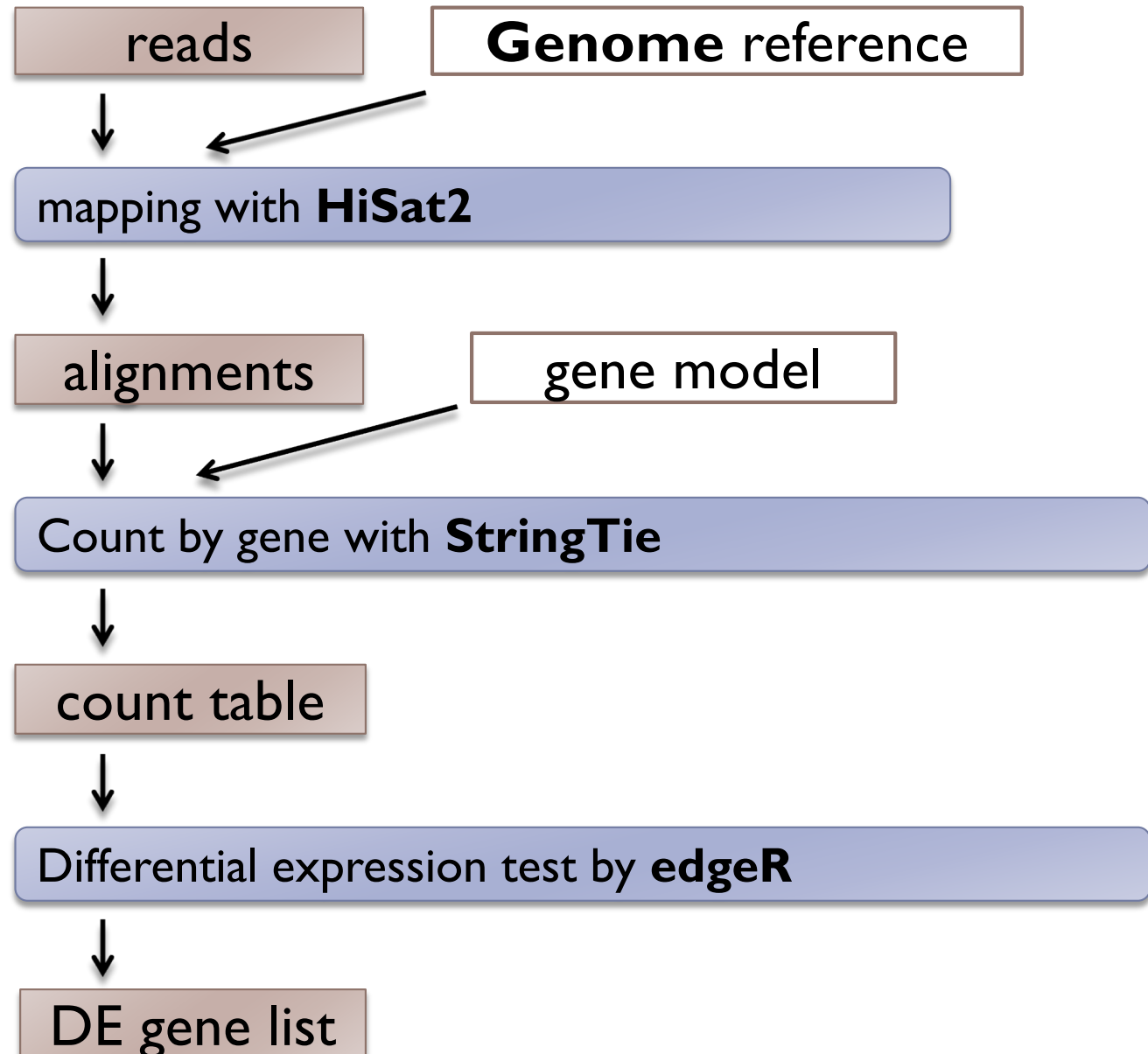
- ▶ **Genome** – standard for genome-known species
- ▶ **Transcript** – the only way for genome-unknown species
 - can be used for genome-known species

トランスクリプトーム参照型

Common workflow



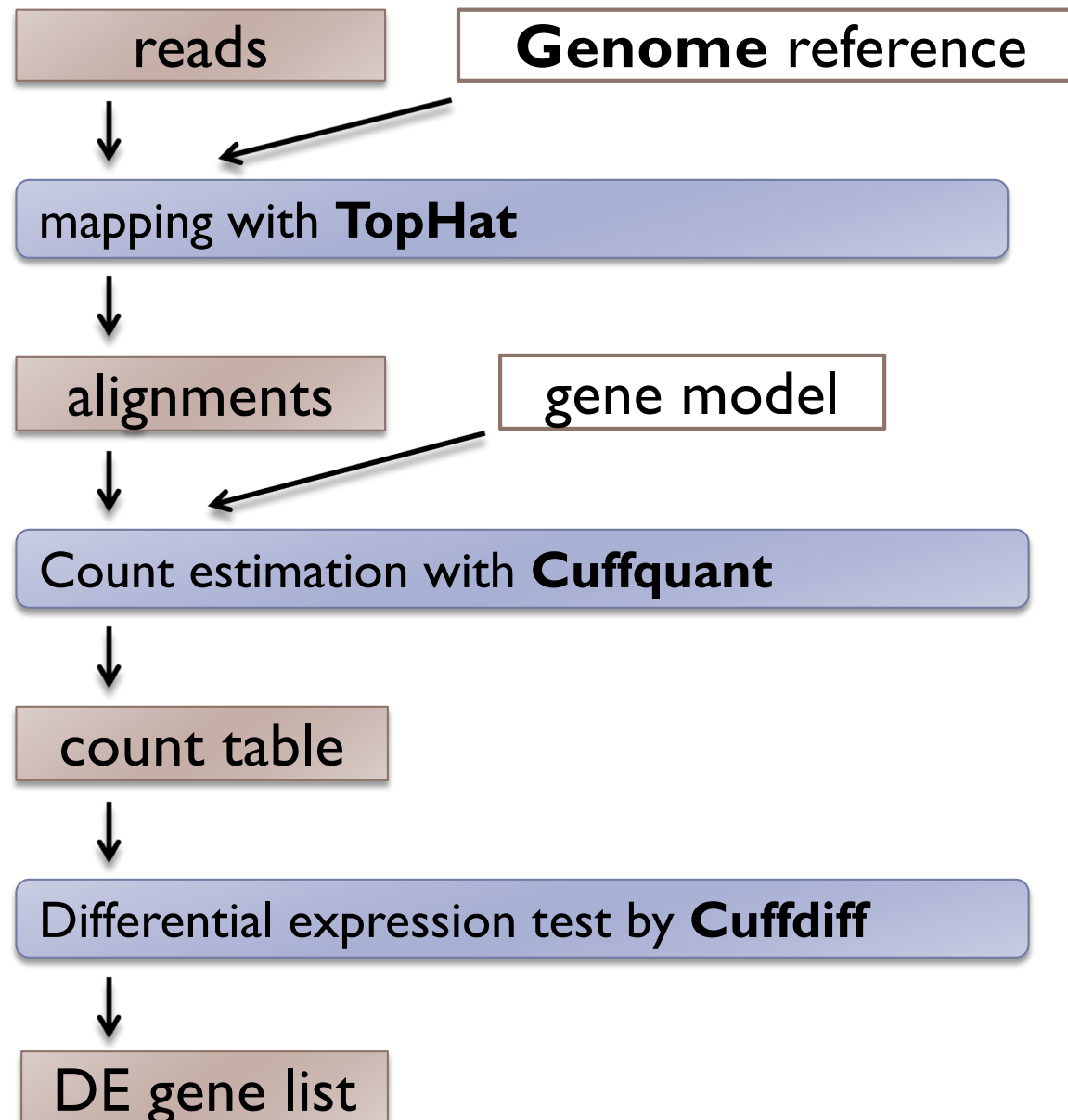
A genome-based pipeline



New “Tuxedo pipeline”

Pertea et al. 2016 Nat
Protoc

A genome-based pipeline

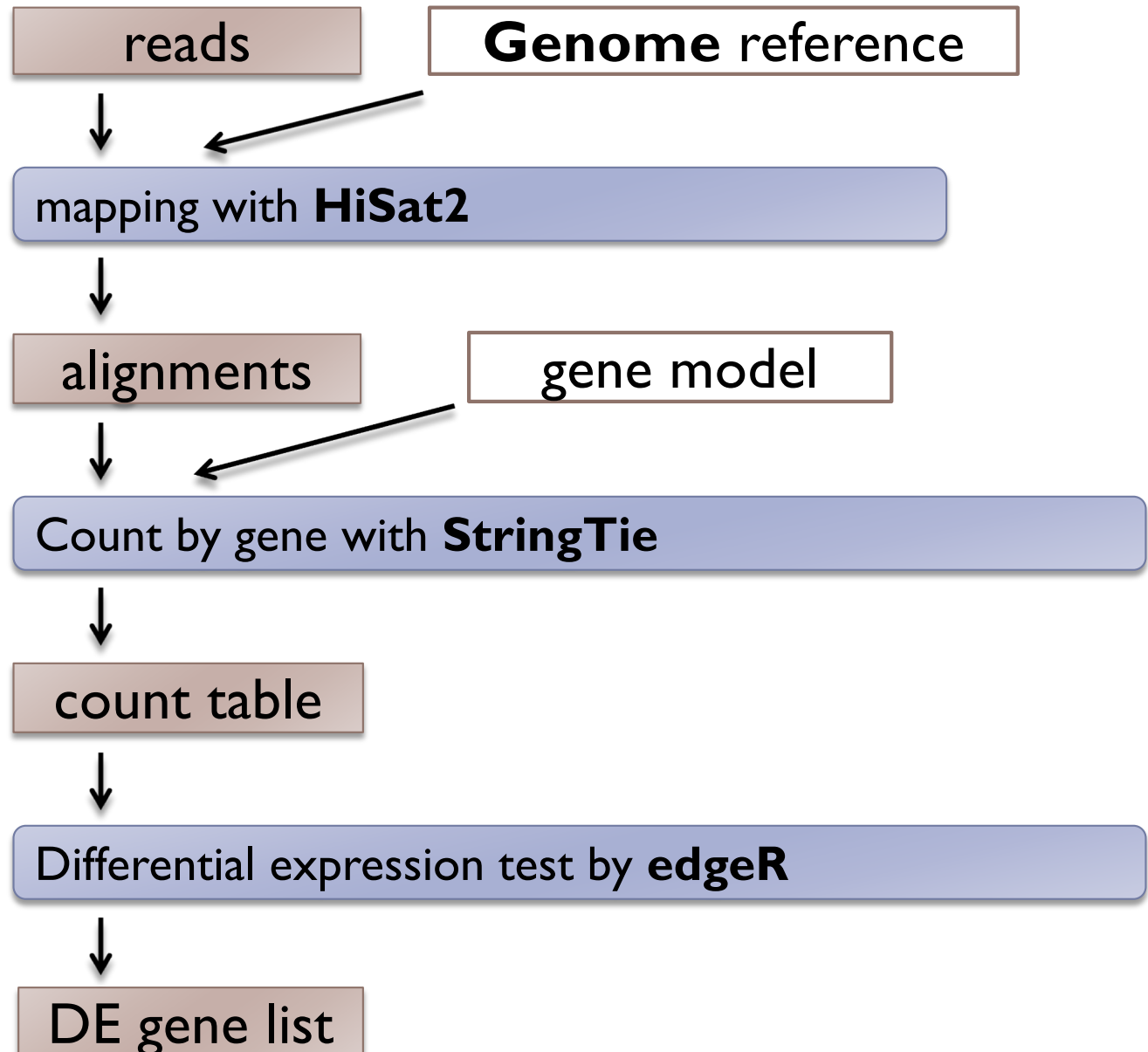


“Tuxedo pipeline”

A genome-based pipeline

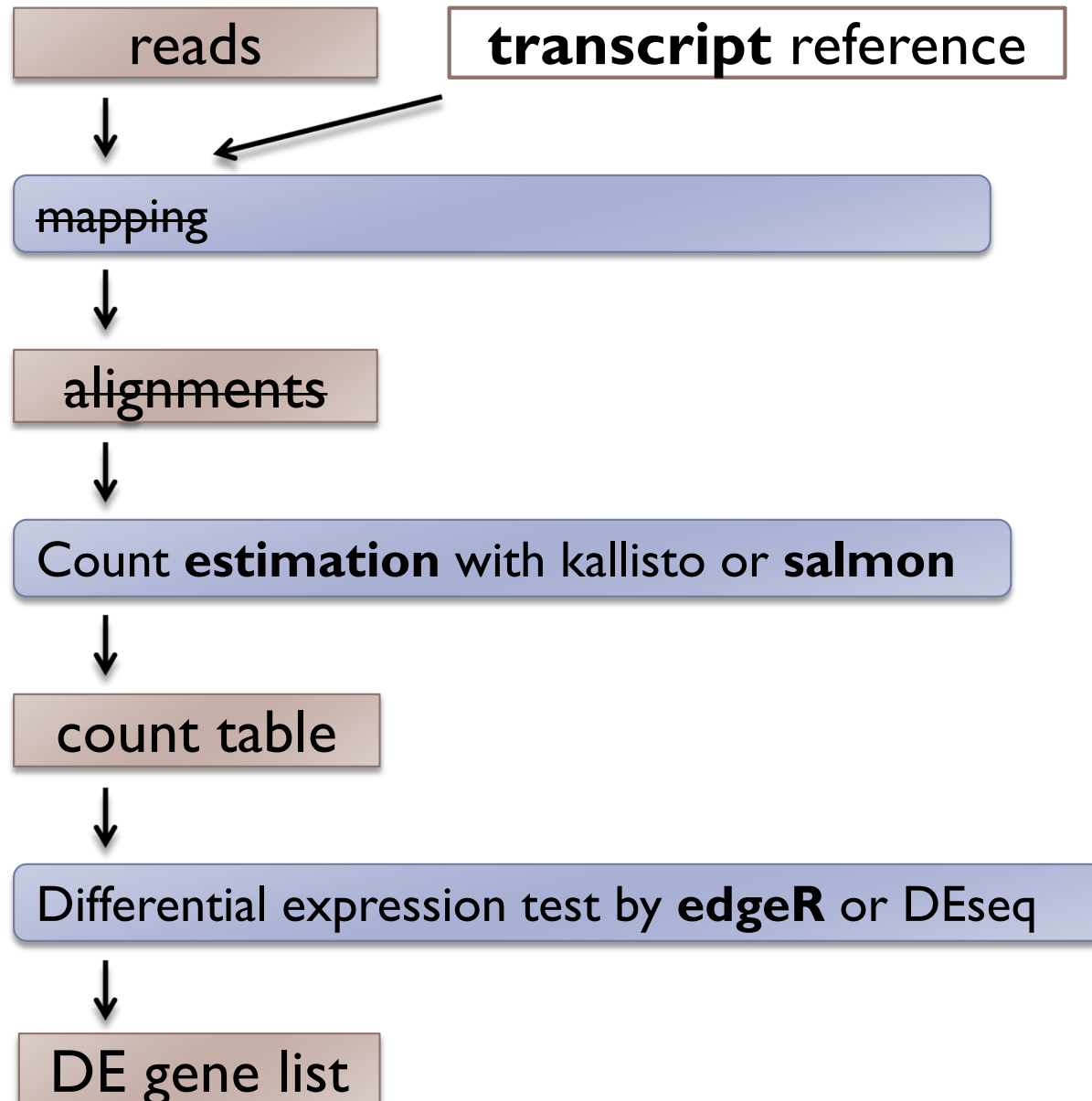
other popular tools

- STAR
- Subread
- HTSeq count
- featureCounts



A transcript-based pipeline (alignment-free method)

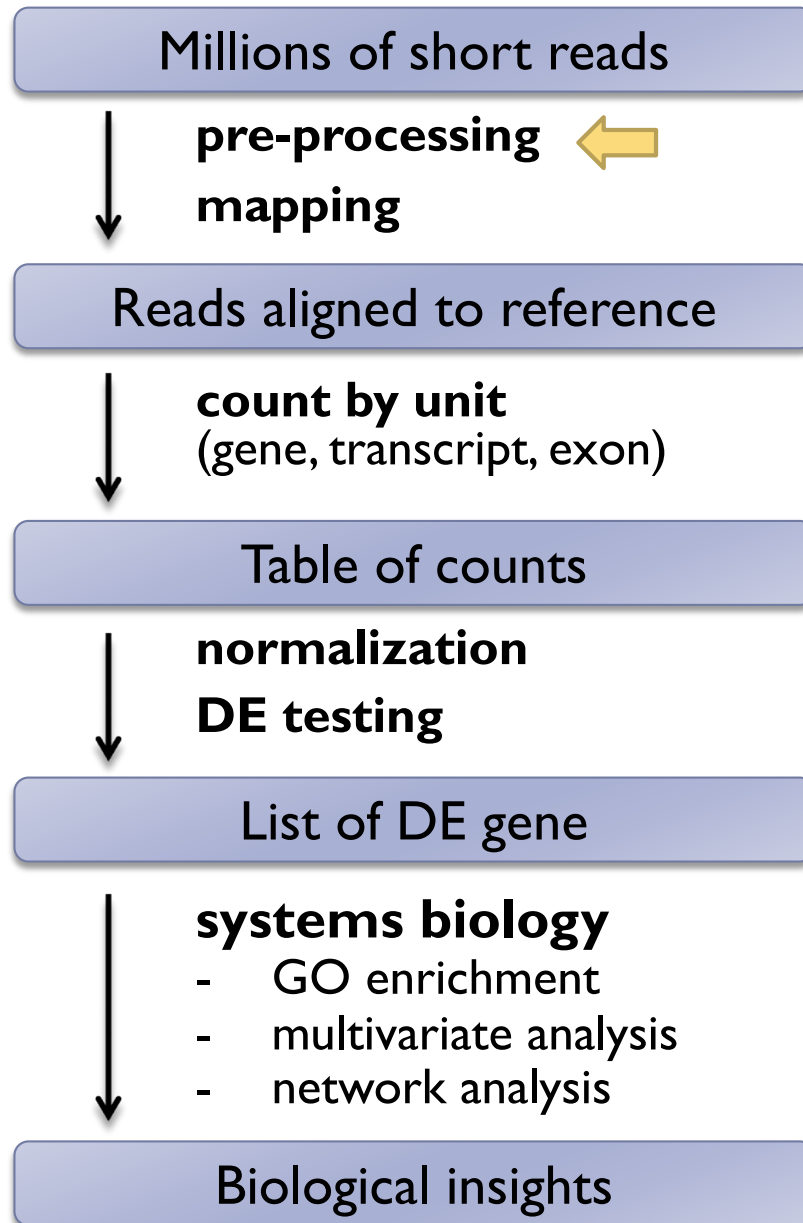
トランスクリプトーム参照型



	ゲノム参照型	トランスクリプト参照型
必要なリファレンス	ゲノム配列 (FASTA) + 遺伝子アノテーション (GFF/GTF)	トランスクリプト配列 (FASTA) のみ
代表的ツール	HiSat2, STAR, StringTie, featureCounts	Salmon, Kallisto, RSEM
スプライシング検出	可能	基本的に不可 (参照トランスクリプトに依存)
新規遺伝子の発見	可能	不可 (既知トランスクリプトに限定)
計算速度	遅い	極めて高速
メモリ・計算資源	多く必要	軽量
解析の網羅性	高い (未知領域を含めた解析が可能)	既知のトランスクリプト配列に限定

RNA-seq analysis pipeline for DE

Differential Expression analysis



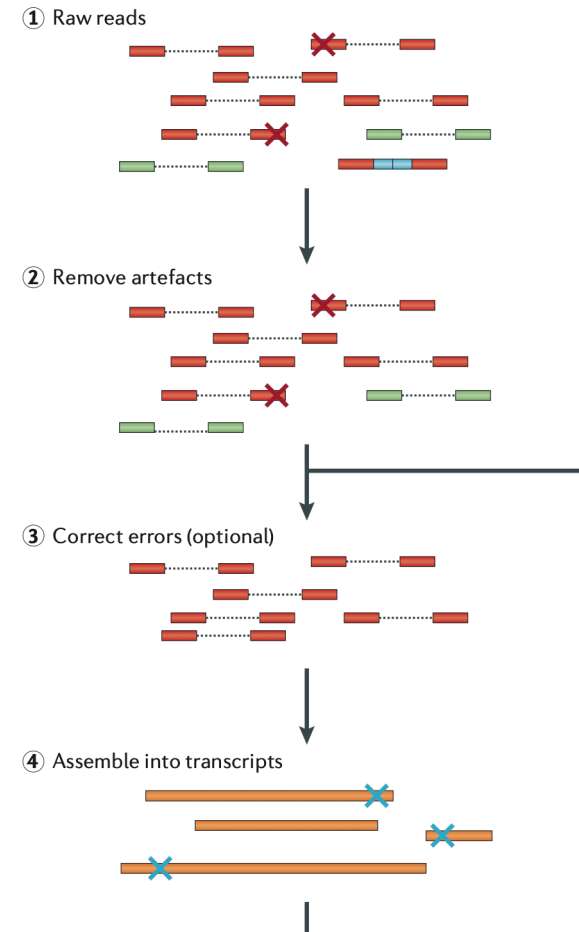
Read QC and Pre-processing

► Read QC

- Tools: FastQC etc.

► Pre-processing

- Filter or trim by base quality
- Remove artifacts
 - adaptors
 - low complexity reads
 - PCR duplications (optional)
- Remove rRNA and other contaminations (optional)
- Sequence error correction (optional)
- Tools: `cutadapt`, `trimmomatic`, `fastp`, `Trim Galore`



Martin et al (2011) *Nat Rev Genet*

RNA-seq analysis pipeline for DE

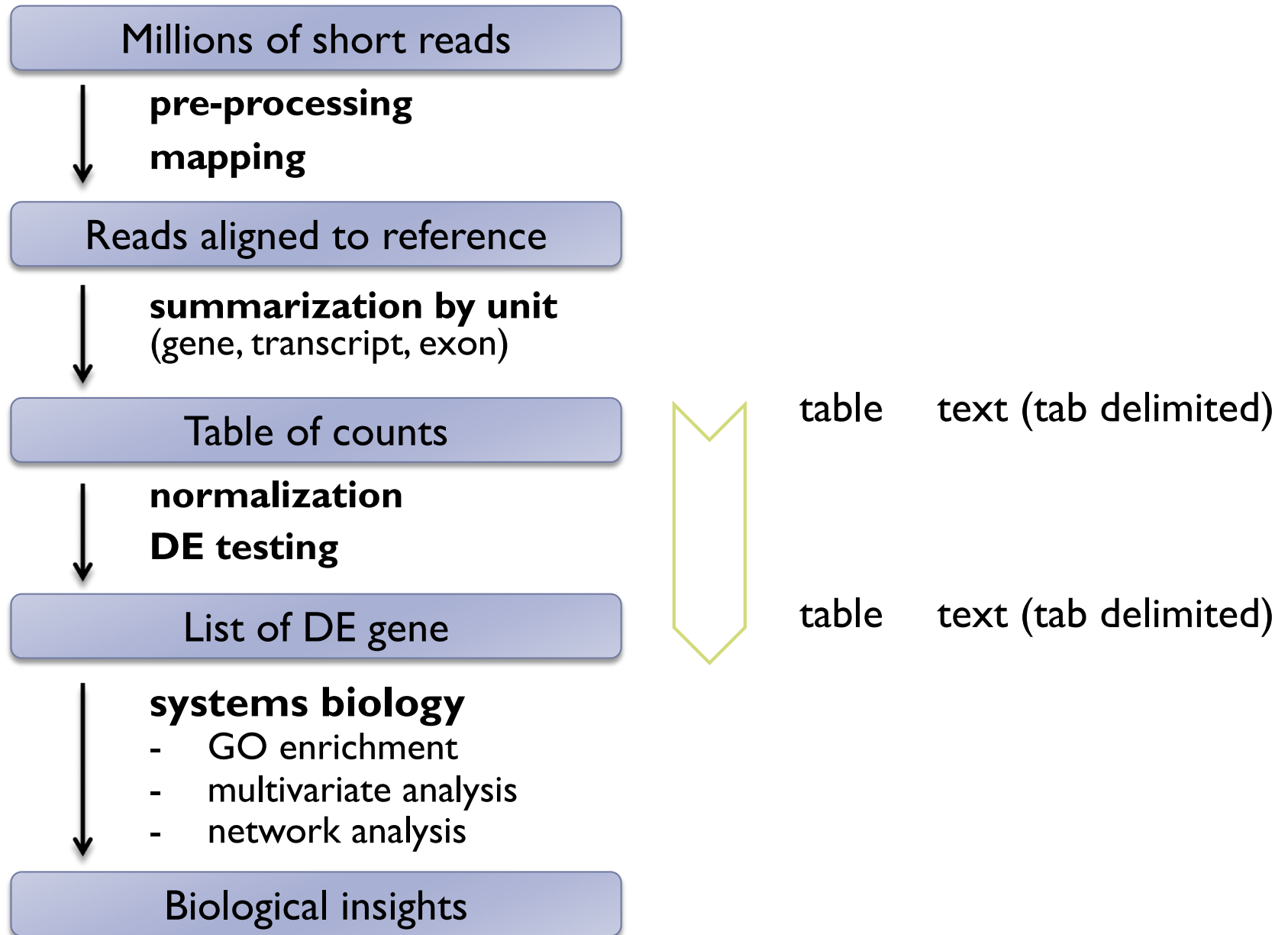


Table of counts



data import

diagnostics

normalization

DE testing

evaluation

List of DE gene

Input

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8	AT1G01070	16	34	6	9	20	1
9	AT1G01080	170	191	382	127	98	184
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14	AT1G01130	9	11	1	0	2	9
15	AT1G01140	181	120	142	161	73	134
16	AT1G01150	0	2	0	0	0	0
17	AT1G01160	117	125	215	86	46	212
18	AT1G01170	74	57	82	36	22	29
19	AT1G01180	46	7	26	24	18	58
20	AT1G01190	0	3	2	1	2	2
21	AT1G01200	5	0	2	0	0	0
22	AT1G01210	178	203	98	205	83	143
23	AT1G01220	26	49	40	21	15	34
24	AT1G01225	4	10	6	6	0	3
25	AT1G01230	72	51	58	70	18	77
26	AT1G01240	81	89	45	62	24	33
27	AT1G01250	1	1	5	1	2	2
28	AT1G01260	15	52	37	33	27	54
29	AT1G01290	7	16	23	30	5	19
30	AT1G01300	75	115	232	89	109	224

Table of counts

Output

	A	B	C	D	E
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5	AT1G51850	-10.07	5.32	2.2E-19	1.4E-15
6	AT2G39380	-9.82	4.97	2.9E-19	1.5E-15
7	AT2G44370	-10.23	5.48	1.1E-18	4.7E-15
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11	AT1G51820	-9.40	4.40	1.3E-17	3.3E-14
12	AT2G39530	-9.16	4.36	1.7E-17	4.1E-14
13	AT1G40104	-9.58	4.61	2.2E-17	4.9E-14
14	AT2G30750	-9.97	4.64	1.5E-16	3.0E-13
15	AT2G17740	-9.12	4.18	1.8E-16	3.3E-13
16	AT2G07981	-10.07	4.75	1.0E-15	1.8E-12
17	AT1G30700	-9.21	3.91	3.5E-15	5.8E-12
18	AT5G31770	-9.12	3.76	1.9E-14	2.7E-11
19	AT1G51800	-8.47	3.99	1.9E-14	2.7E-11
20	AT1G30720	-9.06	3.75	4.7E-14	6.5E-11
21	AT4G12500	-6.58	4.39	1.5E-13	1.9E-10
22	AT1G07160	-10.76	5.02	2.0E-13	2.5E-10
23	AT2G39200	-7.57	3.98	6.0E-13	7.2E-10
24	AT2G10600	-9.30	3.50	8.8E-13	1.0E-09
25	AT1G65500	-9.60	3.50	9.6E-13	1.0E-09
26	AT5G39580	-9.24	3.45	1.3E-12	1.3E-09
27	AT5G44380	-9.51	3.45	1.7E-12	1.7E-09
28	AT5G13220	-7.20	3.88	2.8E-12	2.7E-09
29	AT2G45220	-8.45	3.54	3.9E-12	3.6E-09
30	AT1G30730	-9.80	3.45	6.0E-12	5.4E-09
31	AT1G64170	-9.91	3.39	1.2E-11	1.1E-08
32	AT4G20860	-8.10	3.49	1.5E-11	1.2E-08
33	AT4G23550	-10.08	3.49	1.5E-11	1.2E-08
34	AT4G12490	-6.75	3.88	1.6E-11	1.2E-08
35	AT5G64120	-7.09	3.76	2.1E-11	1.6E-08

List of DE genes

Table of significance score

Identify differentially expressed genes (DEG)

Question: Which are differentially expressed genes (DEG)?

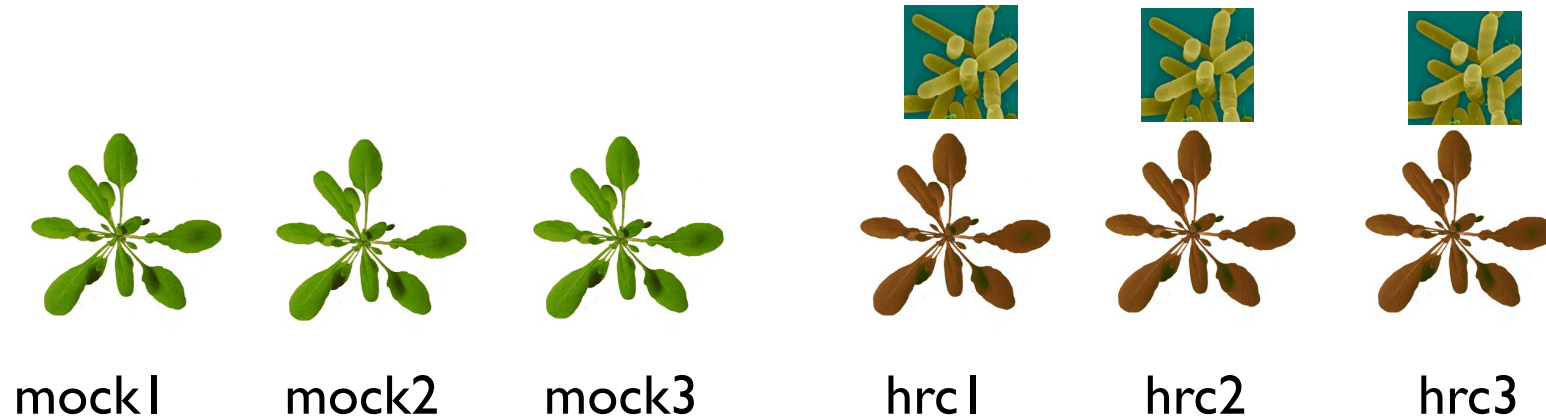
[simple examples (pairwise comparison)]

- mutant v.s. WT
- tissue A v.s. tissue B
- developmental time point A (ex. Early) v.s. B (ex. Late)

Goal:

- Find DE genes
- Rank by significance

Example: Arabidopsis RNA-seq



mock inoculation (treated w/
10mM MgCl₂)

Challenged by defense-eliciting delta-
hrcC mutant of *Pseudomonas syringae*
pathovar *tomato* DC3000.

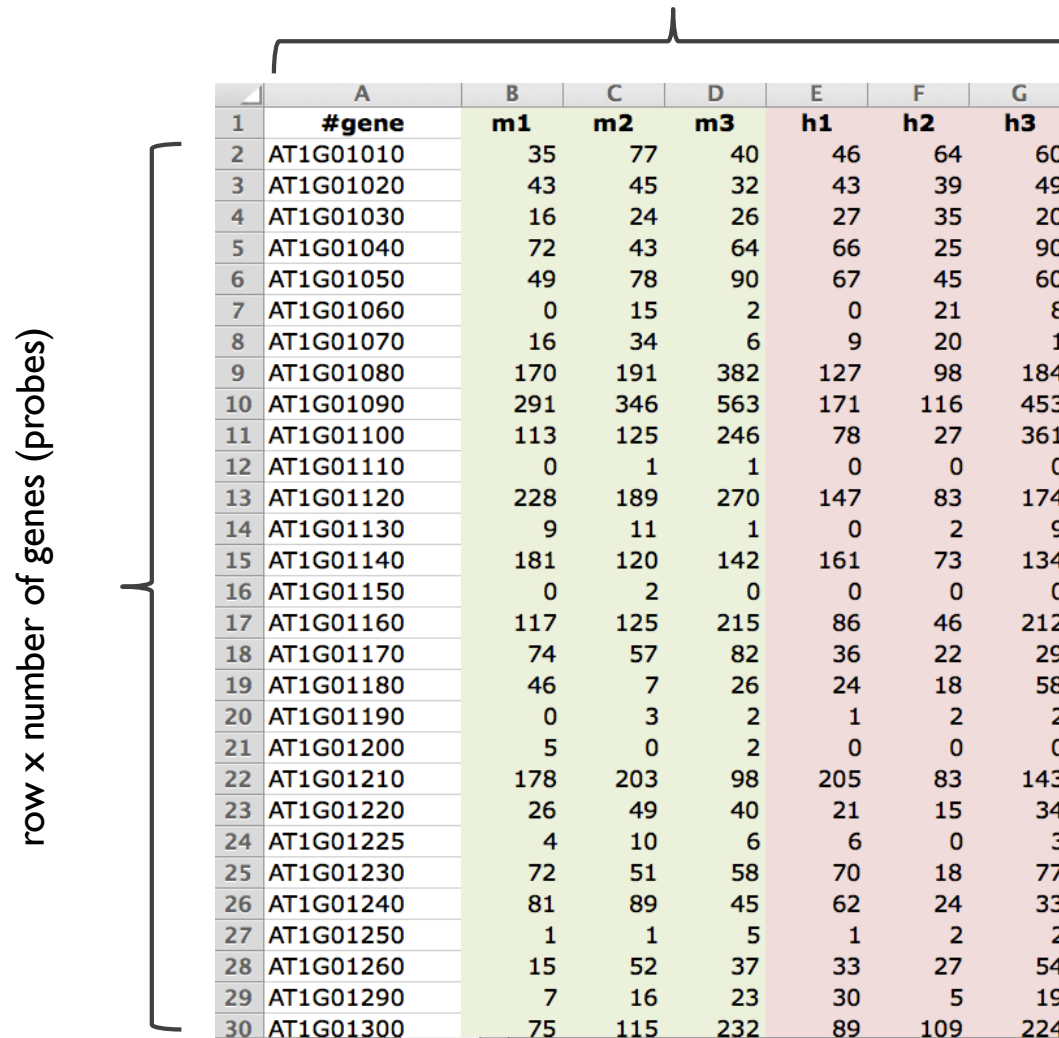
- ▶ 6 libraries = 2 groups x 3 biological replicates

Di, Y. et al. *Stat Appl Genet Mol* (2011).
Cumbie, J. S. et al. *PLoS ONE* (2011).

Input

- ▶ Typical primary data = matrix of #genes x #samples

column x number of samples (libraries)



The diagram illustrates a matrix of gene expression data. A vertical bracket on the left side of the table is labeled "row x number of genes (probes)". A horizontal bracket above the table is labeled "column x number of samples (libraries)". The table itself has 30 rows and 8 columns. The first column contains row numbers 1 through 30. The second column contains gene identifiers starting with "AT1G". The remaining seven columns are labeled with sample identifiers: m1, m2, m3, h1, h2, and h3. The data values are numerical, representing expression levels.

	A	B	C	D	E	F	G
1	#gene	m1	m2	m3	h1	h2	h3
2	AT1G01010	35	77	40	46	64	60
3	AT1G01020	43	45	32	43	39	49
4	AT1G01030	16	24	26	27	35	20
5	AT1G01040	72	43	64	66	25	90
6	AT1G01050	49	78	90	67	45	60
7	AT1G01060	0	15	2	0	21	8
8	AT1G01070	16	34	6	9	20	1
9	AT1G01080	170	191	382	127	98	184
10	AT1G01090	291	346	563	171	116	453
11	AT1G01100	113	125	246	78	27	361
12	AT1G01110	0	1	1	0	0	0
13	AT1G01120	228	189	270	147	83	174
14	AT1G01130	9	11	1	0	2	9
15	AT1G01140	181	120	142	161	73	134
16	AT1G01150	0	2	0	0	0	0
17	AT1G01160	117	125	215	86	46	212
18	AT1G01170	74	57	82	36	22	29
19	AT1G01180	46	7	26	24	18	58
20	AT1G01190	0	3	2	1	2	2
21	AT1G01200	5	0	2	0	0	0
22	AT1G01210	178	203	98	205	83	143
23	AT1G01220	26	49	40	21	15	34
24	AT1G01225	4	10	6	6	0	3
25	AT1G01230	72	51	58	70	18	77
26	AT1G01240	81	89	45	62	24	33
27	AT1G01250	1	1	5	1	2	2
28	AT1G01260	15	52	37	33	27	54
29	AT1G01290	7	16	23	30	5	19
30	AT1G01300	75	115	232	89	109	224

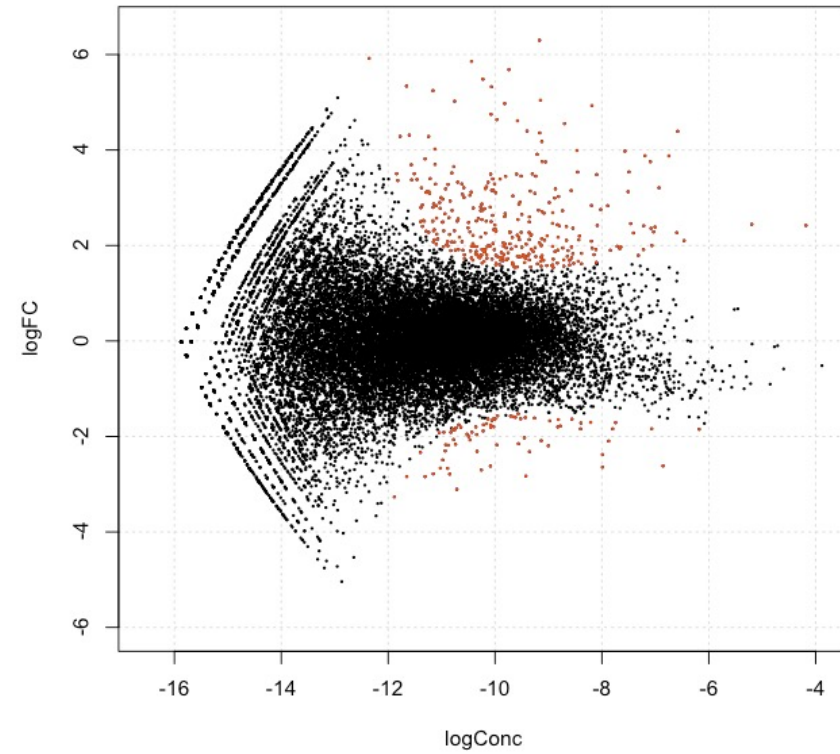
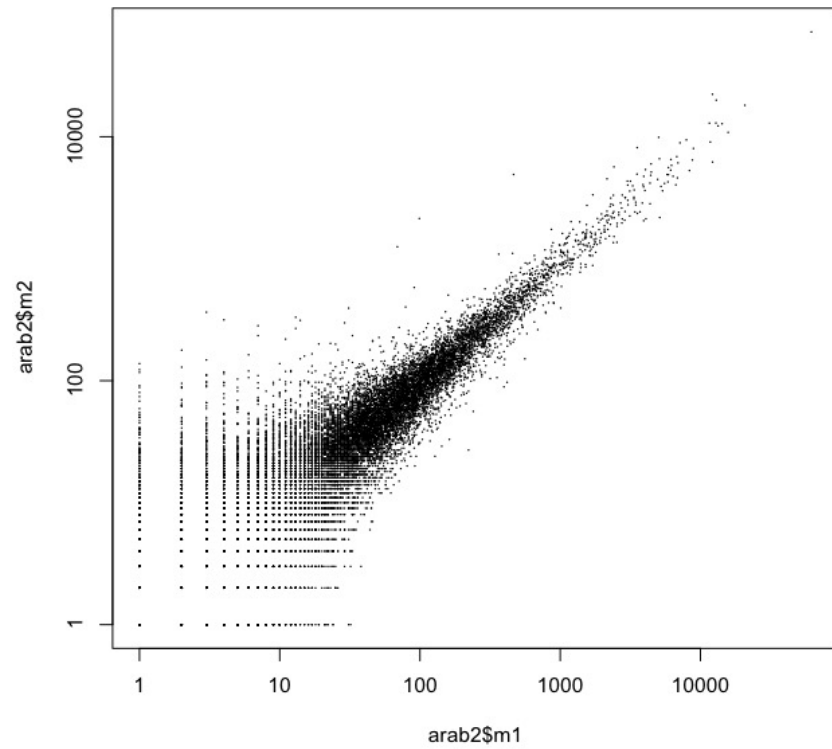
Import count table / diagnostics

Look into the input data first

- ▶ Quick view of the table (tools: R, MS Excel etc.)
 - ▶ Check: Format, data structure, data size etc.
- ▶ Scatter plot, MA plot (tools: R, MS Excel etc.)

H11							
	A	B	C	D	E	F	G
1	#gene	m1	m2	m3	h1	h2	h3
2	AT1G01010	35	77	40	46	64	60
3	AT1G01020	43	45	32	43	39	49
4	AT1G01030	16	24	26	27	35	20
5	AT1G01040	72	43	64	66	25	90
6	AT1G01050	49	78	90	67	45	60
7	AT1G01060	0	15	2	0	21	8
8	AT1G01070	16	34	6	9	20	1
9	AT1G01080	170	191	382	127	98	184
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12	AT1G01110	0	1	1	0	0	0
13	AT1G01120	228	189	270	147	83	174
14	AT1G01130	9	11	1	0	2	9
15	AT1G01140	181	120	142	161	73	134
16	AT1G01150	0	2	0	0	0	0
17	AT1G01160	117	125	215	86	46	212
18	AT1G01170	74	57	82	36	22	29
19	AT1G01180	46	7	26	24	18	58
20	AT1G01190	0	3	2	1	2	2
21	AT1G01200	5	0	2	0	0	0
22	AT1G01210	178	203	98	205	83	143
23	AT1G01220	26	49	40	21	15	34
24	AT1G01225	4	10	6	6	0	3
25	AT1G01230	72	51	58	70	18	77
26	AT1G01240	81	89	45	62	24	33
27	AT1G01250	1	1	5	1	2	2
28	AT1G01260	15	52	37	33	27	54
29	AT1G01290	7	16	23	30	5	19
30	AT1G01300	75	115	232	89	109	224
31	AT1G01310	0	3	0	0	3	0
32	AT1G01320	449	590	1421	483	291	966

Diagnostics: Scatter plot & MA plot



Let's try: data import and quick check

```
> dat <- read.delim("arab2.txt", row.names=1) # ファイルパスは各自の環境に合わせて
> head(dat)                                # look at the first several lines
      m1 m2 m3 h1 h2 h3                    # for checking
AT1G01010 35 77 40 46 64 60
AT1G01020 43 45 32 43 39 49
AT1G01030 16 24 26 27 35 20
AT1G01040 72 43 64 66 25 90
AT1G01050 49 78 90 67 45 60
AT1G01060  0 15  2  0 21  8

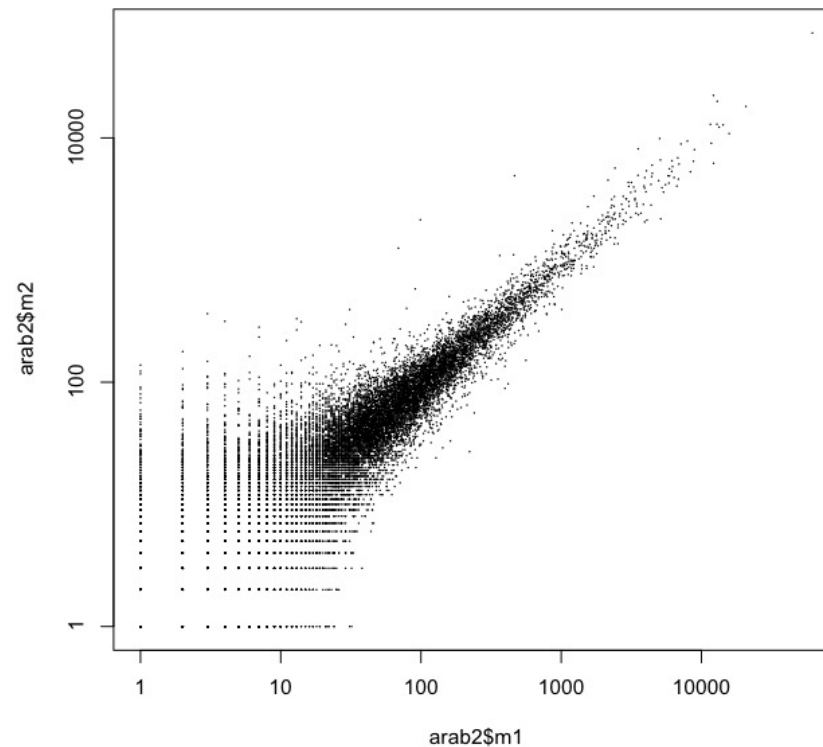
> dim(dat)                                # get numbers of rows and columns
[1] 26221      6

> colSums(dat)                            # get column sums
      m1      m2      m3      h1      h2      h3
1902032 1934029 3259705 2129854 1295304 3526579
```

演習問題 ex301

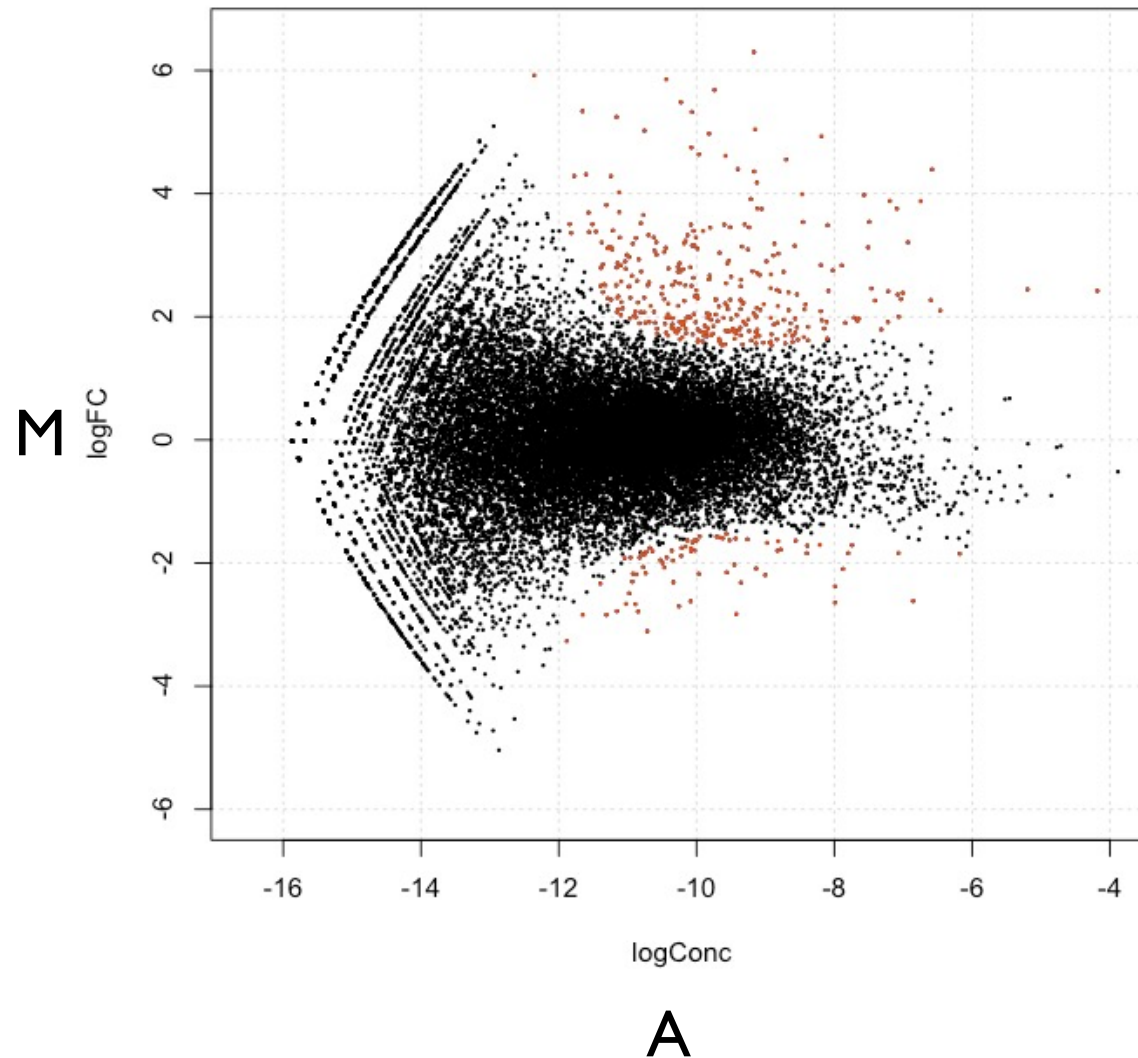
Let's try: Scatter plot

```
> plot(dat$m1 + 1, dat$m2 + 1, log="xy")
```



演習問題 ex301

MA plot



M: log fold-change

A: log intensity average

$$M = \log_2(R/G) = \log_2(R) - \log_2(G)$$

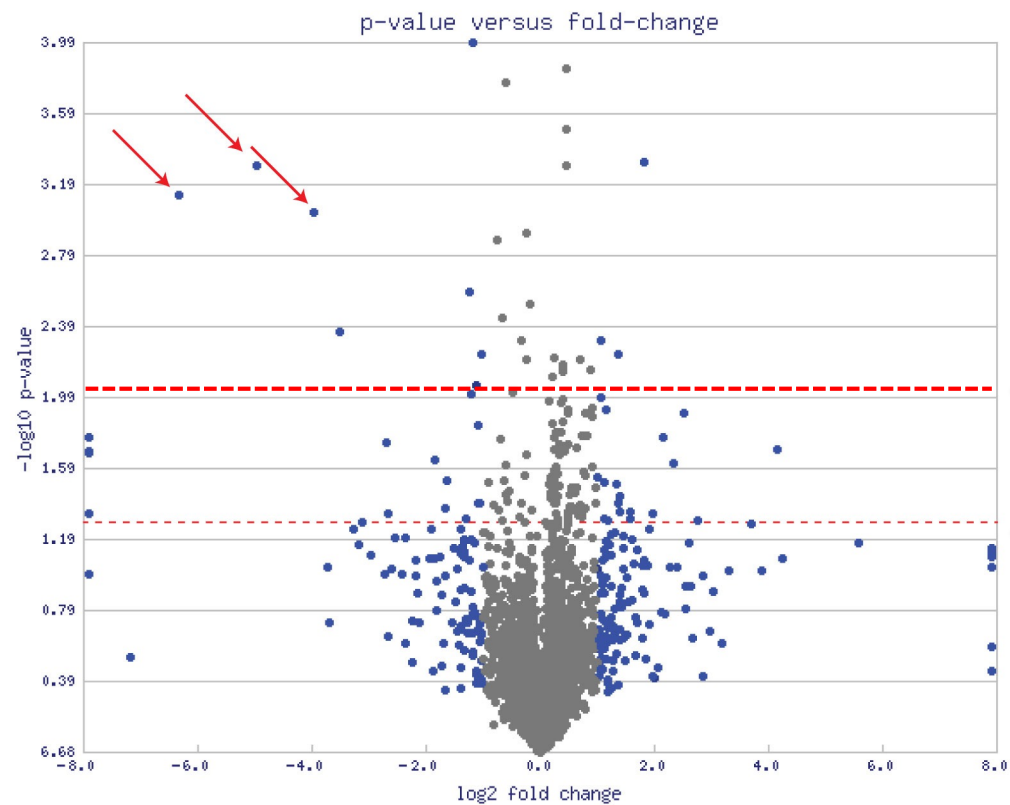
$$A = \frac{1}{2}\log_2(RG) = \frac{1}{2}(\log_2(R) + \log_2(G))$$

R: expression level of sample 1

G: expression level of sample 2

演習問題 ex302

Volcano Plot



$P = 0.01$
 $(-\log_{10}(0.01) = 2)$

$P = 0.05$
 $(-\log_{10}(0.05) = 1.301)$

Fig:Wikipedia

- ▶ X axis: log fold change – 発現比
- ▶ Y axis: log p-value -- significance

演習問題 ex303

DEG解析を学んだ後でtry

Normalization

- ▶ What is normalization? Why is it required?
- ▶ Types of normalization.
- ▶ RNA-seq specific issue.

Normalization

What is normalization? Why is it required?

- ▶ Normalization means to adjust transcriptome data for effects which arise from variation in the technology rather than from biological differences between the RNA samples or between genes.
- ▶ Normalization is an essential step in the analysis of DE from RNA-seq data to make them really comparable.

Normalization: two types

▶ Between-libraries

ライブラリ間補正

- ▶ Comparing expression (counts) of genes between libraries

▶ Within-library

ライブラリ内補正

- ▶ Comparing expression (counts) of genes within a library
(should be possible with NGS – in contrast to microarray)

Normalization

- ▶ **Between-library:**

gene vs gene **between** libraries/sample

Adjust by the total number of reads

- ▶ CPM (Counts Per Million mapped reads)

$$\text{CPM}_i = \frac{X_i}{\frac{N}{10^6}} = \frac{X_i}{N} \cdot 10^6$$

RPM (Read Per Million mapped reads) と呼ばれる

X_i : count of gene

N : number of fragments sequenced

Normalization

- ▶ **Within-library:**

gene vs gene **within** sample

Longer transcripts gets higher counts. => Needs normalization by length

- ▶ RPKM/FPKM (Reads/Fragments Per Kb per Million mapped reads)

$$\text{FPKM}_i = \frac{X_i}{\left(\frac{\tilde{l}_i}{10^3}\right) \left(\frac{N}{10^6}\right)} = \frac{X_i}{\tilde{l}_i N} \cdot 10^9$$

- ▶ TPM (Transcript per million)

$$\text{TPM}_i = \frac{X_i}{\tilde{l}_i} \cdot \left(\frac{1}{\sum_j \frac{X_j}{\tilde{l}_j}} \right) \cdot 10^6$$

$\tilde{l}_i$: effective length of gene

N : number of fragments sequenced

X_i : count of gene

► Relationship between TPM and FPKM

$$\text{TPM}_i = \left(\frac{\text{FPKM}_i}{\sum_j \text{FPKM}_j} \right) \cdot 10^6$$

```
countToTpm <- function(counts, effLen){  
  rate <- log(counts) - log(effLen)  
  denom <- log(sum(exp(rate)))  
  exp(rate - denom + log(1e6))  
}  
  
countToFpkm <- function(counts, effLen){  
  N <- sum(counts)  
  exp( log(counts) + log(1e9) - log(effLen) - log(N) )  
}  
  
fpkmToTpm <- function(fpkm){  
  exp(log(fpkm) - log(sum(fpkm)) + log(1e6))  
}  
  
countToEffCounts <- function(counts, len, effLen){  
  counts * (len / effLen)  
}
```

More robust normalization

- ▶ **Simple scaling**

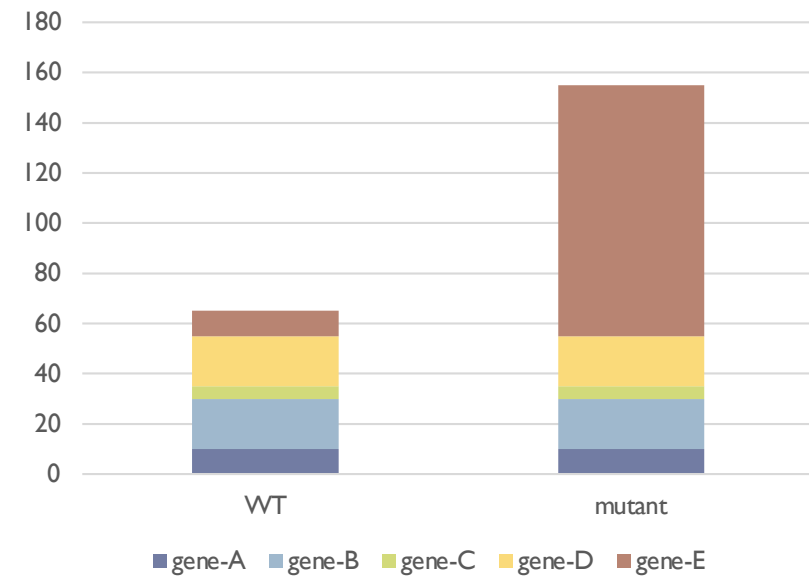
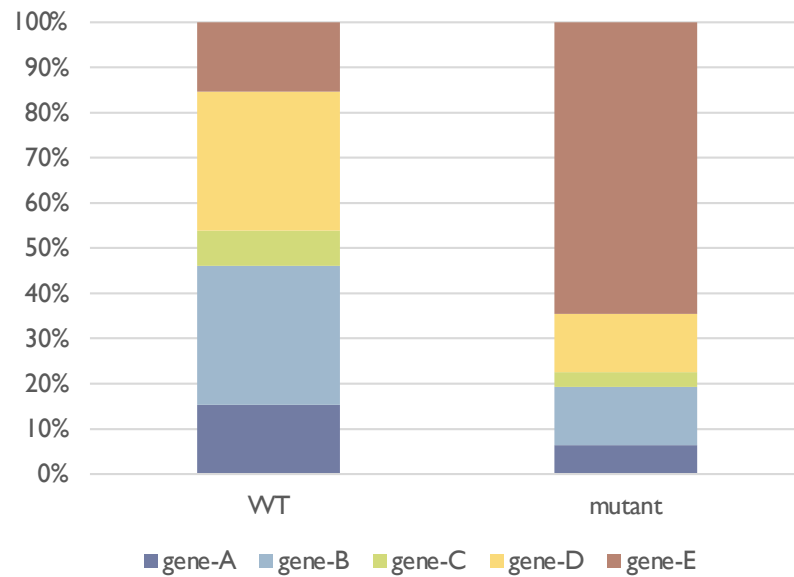
- ▶ Simple scaling such as CPM, TPM or RPKM/FPKM works well in many cases. But RNA comparison differences requires further normalization.

- ▶ **Composition effects**

- ▶ A small number of highly expressed genes can consume a significant amount of the total sequence.
- ▶ Strategies to control composition effects
 - ▶ estimate scaling factors from data and statistical models
 - ▶ quantile normalization
 - ▶ ...

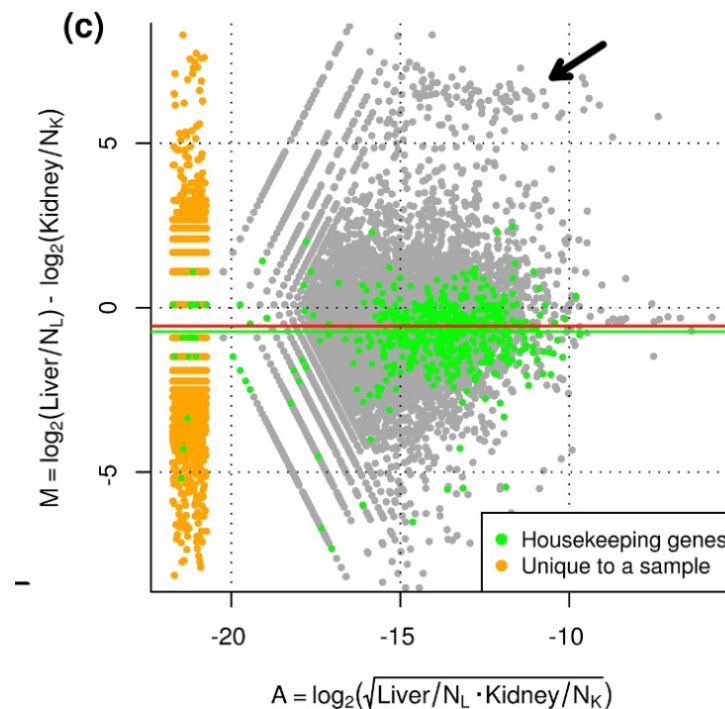
Composition effect

	WT	mutant
gene-A	10	10
gene-B	20	20
gene-C	5	5
gene-D	20	20
gene-E	10	100



TMM Method

- ▶ Trimmed Mean of M values method (Robinson et al., 2010)



1. Calculate M and A for all genes
2. Remove genes with M of top and bottom 30%
3. Remove genes with A of top and bottom 5%
4. Calculate scaling factor using remained genes.

Implementation

edgeR

- ▶ **Model:** An over dispersed Poisson model, **negative binomial (NB) model** is used
- ▶ **Normalization: TMM method** (trimmed mean of M values; Robinson et al., 2010), RLE (Anders et al., 2010) and upperquantile (Bullard et al., 2010)

edgeRの演習は次の
セッションにて

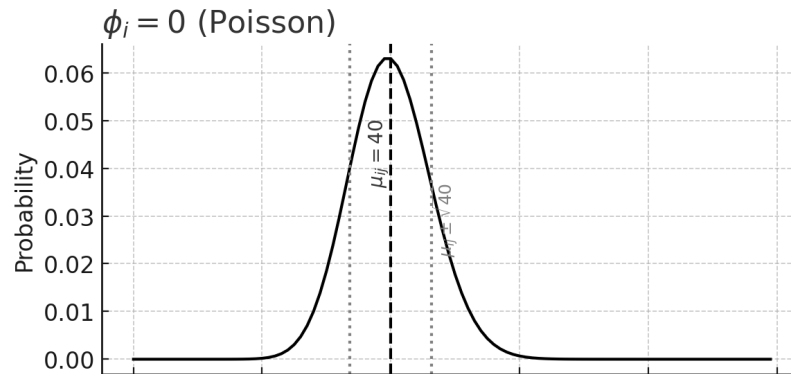
Statistical issues in RNA-Seq DEG analysis

- ▶ RNA-seq count data is Non-Gaussian
- ▶ N (biological replicates) is so small
- ▶ Multiple comparisons (多重検定の問題)

RNA-seq data is Non-Gaussian

- ▶ RNA-seq data
 - ▶ Discrete-valued data (離散値)
 - ▶ Not normally distributed random variables
 - ▶ **Poisson distribution** for technical replicates
 - ▶ **Negative binomial distribution** for biological replicates.
(負の二項分布)

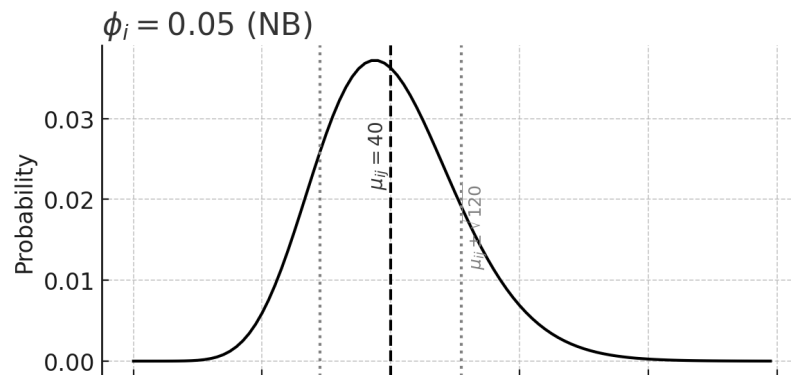
Negative binomial distribution (負の二項分布)



遺伝子*i*のサンプル*j*におけるリード数 X_{ij} が負の二項分布に従うとすると、

$$X_{ij} \sim \text{NB}(\mu_{ij}, \phi_i)$$

μ_{ij} はその遺伝子の期待値(平均発現量)、 ϕ_i は分散を制御するパラメータ(dispersion parameter)

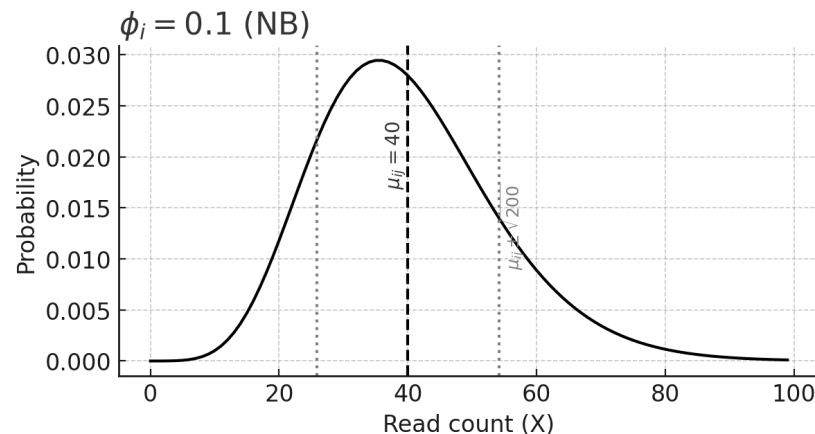


確立密度関数は以下のように定義される:

$$P(X_{ij} = k) = \frac{\Gamma(k + \frac{1}{\phi_i})}{k! \Gamma(\frac{1}{\phi_i})} \left(\frac{1}{1 + \phi_i \mu_{ij}} \right)^{\frac{1}{\phi_i}} \left(\frac{\phi_i \mu_{ij}}{1 + \phi_i \mu_{ij}} \right)^k \quad (k = 0, 1, 2, \dots)$$

平均と分散は以下のように定義される:

$$E[X_{ij}] = \mu_{ij}, \quad \text{Var}[X_{ij}] = \mu_{ij} + \phi_i \mu_{ij}^2$$



Example of pairwise DE using edgeR

```
library(edgeR) # edgeR ライブラリを読み込み

dat <- read.delim("count_data.txt", row.names=1) # カウントテーブルを読み込み
# (オプション) 低発現の遺伝子をフィルタリングする
grp <- c(rep("M", 3), rep("H", 3)) # サンプルを群に割り当て

D <- DGEList(counts=dat, group=grp) # edgeR 用データオブジェクトを作成
D <- calcNormFactors(D) # ライブラリサイズ補正 (TMM法)
D <- estimateDisp(D) # 分散を推定

de <- exactTest(D, pair=c("M", "H")) # 二群間比較 (Exact Test)
topTags(de) # 上位の差次的発現遺伝子を出力
```

(実行例)

演習は次のセクションで

```
Comparison of groups: H-M
      logConc    logFC    P.Value    FDR
AT5G48430 -15.36821  6.255498 9.919041e-12 2.600872e-07
AT5G31702 -15.88641  5.662522 3.637593e-10 4.083773e-06
AT3G55150 -17.01537  5.870635 4.672331e-10 4.083773e-06
...
```

Multiple comparison (多重検定の補正)

- ▶ Adjustment for multiple comparison is essential for RNA-Seq DEG analysis
- ▶ FDR is generally used.

Multi-mapping issue in estimating abundance

- ▶ Source of repeats caused multi-mapping issues in RNA-seq analysis
 - ▶ Isoforms
 - ▶ Very similar paralogs
 - ▶ Transposable elements
- } cannot align reads uniquely
- ▶ Mapping ambiguity should be taken into consideration.



- ▶ Solution
 - ▶ Ignore, split equally, EM algorithm, read coverage-based
 - ▶ RSEM, Slamon, Kallisto: EM

RNA-seq analysis pipeline for DE

